* Case-4 :- $Q = e^{ax} V$, where V is a function of x.

In this case, Eq^{n} (1) reduces to $f(D)\, y = e^{ax} V$

- Let, u be a function of x.

$D(e^{ax} u) = e^{ax} Du + a e^{ax} u$

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$= e^{ax}(D+a)u$

$D^2(e^{ax}u) = D[e^{ax}(D+a)u]$

$= ae^{ax}(D+a)u + e^{ax}(D^2+aD)u$

$= e^{ax}[a(D+a) + D^2+aD]u$

$= e^{ax}[D^2+2aD+a^2]u$

$= e^{ax}(D+a)^2u$

In general, $D^n(e^{ax}u) = e^{ax}(D+a)^n u$

Let, $D^n = f(D)$, $(D+a)^n = f(D+a)$

$f(D)(e^{ax}u) = e^{ax}f(D+a)u$

- Operating both the sides with $\frac{1}{f(D)}$,

$$\frac{1}{f(D)}[f(D)(e^{ax}u)] = \frac{1}{f(D)}[e^{ax}f(D+a)u]$$

$$e^{ax}u = \frac{1}{f(D)}[e^{ax}f(D+a)u]$$

Putting $f(D+a)u = V$,

$$u = \frac{1}{f(D+a)}V$$

$$\therefore e^{ax}\frac{1}{f(D+a)}V = \frac{1}{f(D)}(e^{ax}.V)$$

$\therefore e^{ax}$ Hence, $P.I. = \frac{1}{f(D)}.e^{ax}V$

$$= e^{ax}\frac{1}{f(D+a)}V.$$

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* Examples:-

① Solve $(D+2)^2 y = e^{-2x} \sin x$

→

The A.E. is

$(m+2)^2 = 0$

$\therefore m = -2, -2.$

$\therefore C.F. = e^{-2x}(C_1 + C_2 x)$

$$P.I. = \frac{1}{(D+2)^2} e^{-2x} \sin x$$

$$= e^{-2x} \frac{1}{(D-2+2)^2} \sin x$$

$$= e^{-2x} \frac{1}{D^2} \sin x.$$

$$= e^{-2x} \frac{1}{(-1^2)} \sin x$$

$$= -e^{-2x} \sin x.$$

Hence, the G.S. is $y = e^{-2x}(C_1 + C_2 x) - e^{-2x} \sin x.$

$$= e^{-2x}(C_1 + C_2 x - \sin x).$$

② Solve $(D^2 + 2D + 1) y = \frac{e^{-x}}{x^2}.$

→

The A.E. is $m^2 + 2m + 1 = 0$

$(m+1)^2 = 0$

$\therefore m = -1, -1$

$C.F. = e^{-x}(C_1 + C_2 x)$

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$$P.I. = \frac{1}{D^2+2D+1}\left(\frac{e^{-x}}{x^2}\right)$$

$$= \frac{1}{(D+1)^2}\left(\frac{e^{-x}}{x^2}\right)$$

$$= e^{-x}\frac{1}{(D-1+1)^2}x^{-2}$$

$$= e^{-x}\frac{1}{D^2}x^{-2}$$

$$= e^{-x}\frac{1}{D}\int x^{-2}\,dx$$

$$= e^{-x}\frac{1}{D}\left(\frac{x^{-2+1}}{-2+1}\right)$$

$$= -e^{-x}\frac{1}{D}x^{-1}$$

$$= -e^{-x}\log x$$

Hence, the G.S. is $y = e^{-x}(C_1+C_2x) - e^{-x}\log x$.

$$= e^{-x}(C_1+C_2x-\log x).$$

(3) Solve $(D^2+2D+2)y = e^x\sin x + 7$.

→

The A.E. is $m^2+2m+2=0$

$$\therefore m = \frac{-2\pm\sqrt{4-4(2)}}{2} = \frac{-2\pm\sqrt{-4}}{2} = \frac{-2\pm 2i}{2}$$

$$\therefore m = -1\pm i$$

$$C.F. = e^{-x}(C_1\cos x + C_2\sin x)$$

$$P.I. = \frac{1}{D^2+2D+2}(e^x\sin x + 7)$$

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$$P.I. = \frac{1}{D^2+2D+2} e^x \sin x + \frac{1}{D^2+2D+2} 7e^{0x}$$

$$= e^x \frac{1}{(D+1)^2+2(D+1)+2} \sin x + \frac{1}{0^2+2(0)+2} 7$$

$$= e^x \frac{1}{D^2+2D+1+2D+2+2} \sin x + \frac{7}{2}$$

$$= e^x \frac{1}{D^2+4D+5} \sin x + \frac{7}{2}$$

$$= e^x \frac{1}{(-1^2)+4D+5} \sin x + \frac{7}{2}$$

$$= e^x \frac{1}{4D+4} \sin x + \frac{7}{2}$$

$$= \frac{e^x}{4} \frac{1}{D+1} \sin x + \frac{7}{2}$$

$$= \frac{e^x}{4} \frac{1}{D+1} \frac{(D-1)}{(D-1)} \sin x + \frac{7}{2}$$

$$= \frac{e^x}{4} \frac{D-1}{D^2-1} \sin x + \frac{7}{2}$$

$$= \frac{e^x}{4} \frac{(D-1)}{(-1^2)-1} \sin x + \frac{7}{2}$$

$$= \frac{e^x}{4} \frac{(D-1)}{-2} \sin x + \frac{7}{2}$$

$$= -\frac{e^x}{8} (D-1) \sin x + \frac{7}{2}$$

$$= -\frac{e^x}{8} (\cos x - \sin x) + \frac{7}{2}$$

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Hence, the G.S is

$y = e^{-x}(C_1 \cos x + C_2 \sin x) - \frac{1}{8} e^{x}(\cos x - \sin x) + \frac{7}{2}$.

(4) Solve $(D^2 - 2D + 2)y = e^x x^2 + 5 + e^{-2x}$.

→

The A.E. is $m^2 - 2m + 2 = 0$

$$m = \frac{+2 \pm \sqrt{4 - 4(2)}}{2} = \frac{2 \pm \sqrt{-4}}{2} = \frac{2 \pm 2i}{2}$$

$\therefore m = 1 \pm i$

$$C.F. = e^x (C_1 \cos x + C_2 \sin x)$$

$$P.I. = \frac{1}{D^2 - 2D + 2}(e^x x^2 + 5 + e^{-2x})$$

$$= \frac{e^x}{D^2 - 2D + 2} x^2 + \frac{1}{D^2 - 2D + 2} 5e^{0x} + \frac{1}{D^2 - 2D + 2} e^{-2x}$$

$$= e^x \frac{1}{(D+1)^2 - 2(D+1) + 2} x^2 + 5 \frac{1}{0 - 2(0) + 2} e^{0x} + \frac{1}{(-2)^2 - 2(-2) + 2} e^{-2x}$$

$$= e^x \frac{1}{D^2 + 2D + 1 - 2D - 2 + 2} x^2 + \frac{5}{2} + \frac{1}{10} e^{-2x}$$

$$= e^x \frac{1}{D^2 + 1} x^2 + \frac{5}{2} + \frac{1}{10} e^{-2x}.$$

$$= e^x (1 + D^2)^{-1} x^2 + \frac{5}{2} + \frac{1}{10} e^{-2x}.$$

$$= e^x [1 - D^2 + D^4 - \dots] x^2 + \frac{5}{2} + \frac{1}{10} e^{-2x}.$$

$$= e^x [x^2 - D^2 x^2 + D^4 x^2 - \dots] + \frac{5}{2} + \frac{1}{10} e^{-2x}.$$

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$$P.I = e^{x}(x^2 - 2 + 0) + \frac{5}{2} + \frac{1}{10}e^{-2x}$$

$$= e^{x}(x^2 - 2) + \frac{5}{2} + \frac{1}{10}e^{-2x}.$$

Hence, the G.S. is $y = e^{x}(C_1 \cos x + C_2 \sin x) + e^{x}(x^2 - 2) + \frac{5}{2} + \frac{1}{10}e^{-2x}$

(5) Solve $(D^3+1)y = e^{x/2}\sin\left(\frac{\sqrt{3}}{2}x\right)$.

→

The A.E. is $m^3 + 1 = 0$

$(m+1)(m^2 - m + 1) = 0$

$m + 1 = 0, \quad m^2 - m + 1 = 0$

$m = -1, \quad m = \frac{1 \pm \sqrt{1-4}}{2} = \frac{1 \pm \sqrt{3}i}{2} = \frac{1}{2} \pm \frac{\sqrt{3}}{2}i$

$$C.F. = Ce^{-x} + e^{x/2}\left(C_2 \cos\frac{\sqrt{3}}{2}x + C_3 \sin\frac{\sqrt{3}}{2}x\right)$$

$$P.I. = \frac{1}{D^3+1}e^{x/2}\sin\left(\frac{\sqrt{3}}{2}x\right)$$

$$= e^{x/2}\frac{1}{\left[\left(D+\frac{1}{2}\right)^3 + 1\right]}\sin\left(\frac{\sqrt{3}}{2}x\right)$$

$$= e^{x/2}\frac{1}{\left(D^3 + \frac{1}{8} + \frac{3}{2}D^2 + \frac{3}{4}D\right) + 1}\sin\left(\frac{\sqrt{3}}{2}x\right)$$

$$= e^{x/2}\frac{1}{\left(D\cdot D^2 + \frac{1}{8} + \frac{3}{2}D^2 + \frac{3}{4}D\right) + 1}\sin\left(\frac{\sqrt{3}}{2}x\right)$$

$$= e^{x/2}\frac{1}{\left[-\left(\frac{\sqrt{3}}{2}\right)^2 D + \frac{1}{8} + \frac{3}{2}\left\{-\left(\frac{\sqrt{3}}{2}\right)^2\right\} + \frac{3}{4}D\right] + 1}\sin\frac{\sqrt{3}}{2}x$$

$$= 0$$

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$$P.I. = e^{x/2}\left[x\,\frac{1}{3D^2+3D+\frac{3}{4}}\sin\left(\frac{\sqrt{3}}{2}x\right)\right]$$

$$= e^{x/2}\,x\cdot\frac{1}{3\left\{-\left(\frac{\sqrt{3}}{2}\right)^2\right\}+3D+\frac{3}{4}}\sin\left(\frac{\sqrt{3}}{2}x\right)$$

$$= e^{x/2}\cdot x\,\frac{1}{-\frac{9}{4}+3D+\frac{3}{4}}\sin\left(\frac{\sqrt{3}}{2}x\right)$$

$$= e^{x/2}\cdot x\cdot\frac{1}{3D-\frac{3}{2}}\sin\left(\frac{\sqrt{3}}{2}x\right)$$

$$= e^{x/2}\cdot x\cdot\frac{2}{6D-3}\sin\left(\frac{\sqrt{3}}{2}x\right)$$

$$= e^{x/2}\cdot x\,\frac{2}{3}\frac{1}{2D-1}\sin\left(\frac{\sqrt{3}}{2}x\right)$$

$$= e^{x/2}\cdot x\cdot\frac{2}{3}\,\frac{1}{2D-1}\,\frac{2D+1}{2D+1}\sin\left(\frac{\sqrt{3}}{2}x\right)$$

$$= e^{x/2}\cdot x\cdot\frac{2}{3}\,\frac{2D+1}{4D^2-1}\sin\left(\frac{\sqrt{3}}{2}x\right)$$

$$= e^{x/2}\cdot x\cdot\frac{2}{3}\,\frac{2D+1}{4\left\{-\left(\frac{\sqrt{3}}{2}\right)^2\right\}-1}\sin\left(\frac{\sqrt{3}}{2}x\right)$$

$$= e^{x/2}\cdot x\cdot\frac{2}{3}\,\frac{2D+1}{(-4)}\sin\left(\frac{\sqrt{3}}{2}x\right)$$

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$= e^{x/2} \cdot x \cdot \left(\frac{-1}{6}\right)\left[2D\left(\sin\frac{\sqrt{3}}{2}x\right) + \sin\left(\frac{\sqrt{3}}{2}x\right)\right]$

$= \frac{-1}{6} x e^{x/2}\left[8\sqrt{3}\cos\left(\frac{\sqrt{3}}{2}x\right) + \sin\left(\frac{\sqrt{3}}{2}x\right)\right]$

The G.S is

$y = C_1 e^{-x} + e^{x/2}\left(C_2 \cos\frac{\sqrt{3}}{2}x + C_3 \sin\frac{\sqrt{3}}{2}x\right) - \frac{1}{6} x e^{x/2}$

$\left[\sqrt{3}\cos\left(\frac{\sqrt{3}}{2}x\right) + \sin\left(\frac{\sqrt{3}}{2}x\right)\right]$.

* Case V :- $Q = xV$, where V is a function of x.

In this case, eq^{n} (1) reduces to

$f(D)\,y = xV$

- Let, u be a function of x.

$D(xu) = xDu + u$

$D^2(xu) = D(xDu + u)$

$= xD^2u + Du + Du$

$= xD^2u + 2Du$

$D^3(xu) = D(xD^2u + 2Du)$

$= xD^3u + D^2u + 2D^2u$

$= xD^3u + 3D^2u$

In general,

$D^n(xu) = xD^nu + nD^{n-1}u$

$= xD^nu + \left[\frac{d}{dD}(D^n)\right]u$

Let, $D^n = f(D)$

$\therefore f(D)(xu) = x f(D)u + \left[\frac{d}{dD} f(D)\right]u$

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$= x \cdot f(D)u + f'(D)u$

- Putting $f(D)u = V$,

$u = \frac{1}{f(D)} V$ in the above eq^n,

$$f(D)\left[x \cdot \frac{1}{f(D)} V\right] = xV + f'(D)\left[\frac{1}{f(D)} V\right]$$

$$\therefore xV = f(D)\left[x \cdot \frac{1}{f(D)} V\right] - f'(D)\left[\frac{1}{f(D)} V\right]$$

- Operating both the sides with $\frac{1}{f(D)}$

$$\frac{1}{f(D)} xV = \frac{1}{f(D)}\left[f(D)\left(x \frac{1}{f(D)} V\right)\right] - \frac{1}{f(D)}\left[f'(D)\left(\frac{1}{f(D)}\right)V\right]$$

$$= x \frac{1}{f(D)} V - \frac{f'(D)}{[f(D)]^2} V$$

Hence, $P.I = \frac{1}{f(D)} xV$

$$= x \frac{1}{f(D)} V - \frac{f'(D)}{[f(D)]^2} V.$$

* Examples:-

① Solve $(D^2 - 5D + 6)y = x \cdot \cos 2x$

→

The A.E. is $m^2 - 5m + 6 = 0$

$(m-3)(m-2) = 0$

$m = 2, 3$

$\therefore C.F. = C_1 e^{2x} + C_2 e^{3x}$.

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$$P.I. = \frac{1}{D^2-5D+6} x\cdot\cos 2x$$

$$= x\cdot\frac{1}{D^2-5D+6}\cos 2x - \frac{2D-5}{(D^2-5D+6)^2}\cos 2x$$

$$= x\cdot\frac{1}{-2^2-5D+6}\cos 2x - \frac{2D-5}{(-2^2-5D+6)^2}\cos 2x$$

$$= x\cdot\frac{1}{2-5D}\cos 2x - \frac{2D-5}{(2-5D)^2}\cos 2x$$

$$= x\cdot\frac{1}{2-5D}\frac{(2+5D)}{(2+5D)}\cos 2x - \frac{2D-5}{4-20D+25D^2}\cos 2x$$

$$= x\frac{(2+5D)}{4-25D^2}\cos 2x - \frac{2D-5}{4-20D+25(-2^2)}\cos 2x$$

$$= x\cdot\frac{(2+5D)}{4-25(-2^2)}\cos 2x - \frac{2D-5}{4-20D-100}\cos 2x$$

$$= x\cdot\frac{(2+5D)}{104}\cos 2x - \frac{2D-5}{-20D-96}\cos 2x$$

$$= \frac{x}{104}(2\cos 2x + 5D\cos 2x) + \frac{2D-5}{4(5D+24)}\cos 2x$$

$$= \frac{x}{104}(2\cos 2x - 10D\sin 2x) + \frac{2D-5}{4(5D+24)}\frac{(5D-24)}{(5D-24)}\cos 2x$$

$$= \frac{x}{104}(\cos 2x - 5\sin 2x) + \frac{10D^2-48D-25D+120}{4(25D^2-576)}\cos 2x$$

$$= \frac{x}{52}(\cos 2x - 5\sin 2x) + \frac{(10D^2-73D+120)}{4[25(-2^2)-576]}\cos 2x$$

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$$= \frac{x}{52}(\cos 2x - 5\sin 2x) + \frac{(10D^2 - 73D + 120)}{(-2704)} \cos 2x$$

$$= \frac{x}{52}(\cos 2x - 5\sin 2x) - \frac{1}{2704}[-40\cos 2x + 146\sin 2x + 120\cos 2x]$$

$$= \frac{x}{52}(\cos 2x - 5\sin 2x) - \frac{1}{2704}[+80\cos 2x + 146\sin 2x]$$

$$= \frac{x}{52}(\cos 2x - 5\sin 2x) - \frac{1}{1352}(40\cos 2x + 73\sin 2x)$$

Hence, The G.S. is $y = C_1 e^{2x} + C_2 e^{3x} + \frac{x}{52}[\cos 2x - 5\sin 2x] - \frac{1}{1352}[40\cos 2x + 73\sin 2x]$.

(2) Solve $(D^2 - 1)y = xe^x$, where $D = \frac{d}{dx}$.

→

The A.E. is $D^2 - 1 = 0$

$D^2 = 1$

$D = \pm 1$

$\therefore C.F. = C_1 e^x + C_2 e^{-x}$.

$$P.I. = \frac{1}{D^2 - 1} xe^x.$$

$$= e^x \frac{1}{(D+1)^2 - 1} x$$

$$= e^x \frac{1}{D^2 + 2D + 1 - 1} x$$

$$= e^x \frac{1}{D^2 + 2D} x$$

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$$= \frac{e^x}{2D} \frac{1}{\left[1 + \frac{D}{2}\right]} x$$

$$= \frac{e^x}{2D} \left[1 + \frac{D}{2}\right]^{-1} x$$

$$= \frac{e^x}{2D} \left[1 - \frac{D}{2}\right] x$$

$$= \frac{e^x}{2D} \left[x - \frac{1}{2}\right]$$

$$= \frac{e^x}{2} \int \left(x - \frac{1}{2}\right) dx$$

$$= \frac{e^x}{2} \left[\frac{x^2}{2} - \frac{1}{2}x\right]$$

$$= \frac{e^x}{4}(x^2 - x).$$

- Hence, the G.S. is $y = C_1 e^x + C_2 e^{-x} + \frac{e^x}{4}(x^2 - x)$.

③ Solve $(D^2 + 2D + 1)\, y = x e^{-x} \cos x$.

→

The A.E. is $m^2 + 2m + 1 = 0$

$\therefore (m+1)^2 = 0$

$\therefore m = -1, -1.$

$C.F. = (C_1 + C_2 x) e^{-x}.$

$$P.I. = \frac{1}{D^2 + 2D + 1} x e^{-x} \cos x$$

$$= \frac{1}{(D+1)^2} x e^{-x} \cos x$$

$$= e^{-x} \frac{1}{(D-1+1)^2} x \cos x$$

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$= e^{-x} \frac{1}{D^2} x \cos x$

$= e^{-x} \left[\frac{x \cdot 1}{D^2} \cos x - \frac{2D}{(D^2)^2} \cos x \right]$.

$= e^{-x} \left[\frac{x}{-1^2} \cos x - \frac{2D}{(-1^2)^2} \cos x \right]$.

$= e^{-x} (-x \cos x - 2D \cos x)$

$= e^{-x} (-x \cos x + 2 \sin x)$.

Hence, the G.S. is $y = e^{-x}(C_1 + C_2 x) + e^{-x}(-x \cos x + 2 \sin x)$.

(4) Solve $(D^2 - 1)y = x \sin x + e^x + x^2 e^x$.

→

A.E. is $m^2 - 1 = 0$

$m = -1, 1$.

C.F. $= C_1 e^{-x} + C_2 e^x$.

P.I. $= \frac{1}{D^2 - 1} (x \cdot \sin x + e^x + x^2 e^x)$

$= \frac{1}{D^2-1} x \sin x + \frac{1}{D^2-1} e^x + \frac{1}{D^2-1} x^2 e^x$

$= \left[\frac{x \cdot 1}{D^2-1} \sin x - \frac{2D}{(D^2-1)^2} \sin x \right] + x \cdot \frac{1}{2D} e^x + e^x \frac{1}{(D+1)^2 - 1} x^2$

$= x \left[\frac{1}{-1^2 - 1} \sin x - \frac{2D}{(-1^2-1)^2} \sin x \right] + \frac{x}{2} e^x + e^x \frac{1}{D^2 + 2D} x^2$

$= \left[\frac{-x \sin x}{2} - \frac{2D}{4} \sin x \right] + \frac{xe^x}{2} + e^x \frac{1}{2D\left[1 + \frac{D}{2}\right]} x^2$.

$= \left[\frac{-x \sin x}{2} - \frac{\cos x}{2} \right] + \frac{xe^x}{2} + e^x \frac{1}{2D} \left[1 + \frac{D}{2}\right]^{-1} x^2$.

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$$= -\left[\frac{x\sin x}{2} + \frac{\cos x}{2}\right] + \frac{xe^x}{2} + \frac{e^x}{2}\frac{1}{D}\left[1 - \frac{D^2}{2} + \frac{D^2}{4} - \frac{D^3}{8} + \cdots\right] x$$

$$= -\left[\frac{x\sin x}{2} + \frac{\cos x}{2}\right] + \frac{xe^x}{2} + \frac{e^x}{2D}\left[x^2 - x + \frac{1}{2}\right]$$

$$= -\left[\frac{x\sin x}{2} + \frac{\cos x}{2}\right] + \frac{xe^x}{2} + \frac{e^x}{2}\left[\frac{x^3}{3} - \frac{x^2}{2} + \frac{1}{2}x\right]$$

$$= -\left[\frac{x\sin x}{2} + \frac{\cos x}{2}\right] + \frac{1}{2}e^x\left[x + \frac{x^3}{3} - \frac{x^2}{2} + \frac{x}{2}\right]$$

$$= -\left[\frac{x\sin x}{2} + \frac{\cos x}{2}\right] + \frac{1}{2}e^x\left(\frac{x^3}{3} - \frac{x^2}{2} + \frac{3}{2}x\right)$$

Hence, the G.S is

$$y = C_1e^x + C_2e^{-x} - \left[\frac{x\sin x + \cos x}{2}\right] + \frac{1}{2}e^x\left(\frac{x^3}{3} - \frac{x^2}{2} + \frac{3}{2}x\right).$$

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